AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

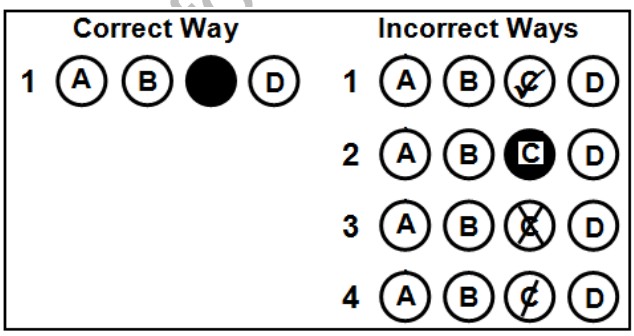
Computer Science Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. Ayan wants to draw some 2D/ 3D illustrations using a computing device.

   The computing device that is BEST to perform the given task is

   A. mouse.
   B. trackball.
   C. smartphone.
   D. graphic tablet.

2. The step of data processing cycle that is involved in taking raw data for processing is

   A. input.
   B. output.
   C. storage.
   D. process.

3. Which of the following characteristics distinguishes a microcomputer from other computer types?

   A. It is physically the smallest type of computer.
   B. It operates without the need for an operating system.
   C. It is primarily used for scientific and research purposes.
   D. It has the highest processing speed among all computers.

4. Consider the given examples of software.

   I. Games
   II. Compiler
   III. Disk Cleaner
   IV. Database Management System

   The examples of system software are

   A. I and II.
   B. I and IV.
   C. II and III.
   D. III and IV.

5. Which type of system software is responsible for performing disk cleanup and defragmentation?

   A. Language translators
   B. Operating system
   C. Utility software
   D. Device driver

6. A type of software that allows users to freely use, modify and distribute the source code is

   A. freeware.
   B. shareware.
   C. licensed software.
   D. open source software.

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7. Example of auto capture device is

   A. mouse.
   B. keyboard.
   C. touch screen.
   D. barcode reader.

8. Which of the following devices can perform both input and output operations?

   A. Printer
   B. Scanner
   C. Light Pen
   D. Touch Screen

9. Scanner scans filled-in bubbles from answer sheets for grading multiple choice questions.

   The device/ reader identified from the given description is a/ an

   A. Optical Mark Reader (OMR).
   B. Optical Character Reader (OCR).
   C. Quick Response (QR) code reader.
   D. Magnetic Ink Character Reader (MICR).

10. Consider the given properties.

   - It is used for converting output for long-term storage and archiving purposes.
   - It is often used for preserving critical documents and records.
   - It has high data security against digital threats.

   The BEST output device that describes the given properties is

   A. Monitor.
   B. Projector.
   C. Flatbed plotter.
   D. Computer Output Microfilm (COM).

11. 4 Mebibyte is equal to

   A. 4,000 bytes.
   B. 4,096 bytes.
   C. 4,000,000 bytes.
   D. 4,194,304 bytes.

12. The characteristic that is shared by both SRAM and DRAM is

   A. use of transistors for data storage.
   B. lack of refresh requirements.
   C. non-volatile storage.
   D. slow access time.

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13. Consider the given characteristics.

I. It was used in early computer systems.
II. It is smaller in size because of no erasure capability.
III. It can retain data for upto 20 years, if not exposed to UV light.

The characteristic(s) that BEST describe PROM is/ are

A. I only.
B. III only.
C. I and II.
D. II and III.

14. The PRIMARY role of the Basic Input/ Output System (BIOS) is

A. connecting to the internet.
B. loading the operating system.
C. executing application software.
D. checking hardware components.

15. In contrast to L2 cache, L1 cache

A. is slower.
B. is fixed in size.
C. consumes less power.
D. is located on the motherboard.

16. Which of the following memories uses optical storage technology?

A. Cache
B. Register
C. CD-ROM
D. USB flash drive

17. The type of storage that is typically used for direct data access and stores data in magnetic form is

A. hard disk.
B. magnetic tapes.
C. USB flash drive.
D. Solid State Drive (SSD).

18. Sana is a video editor. She wants to store a video of 2 GB.

In the given scenario, the type of optical storage media, Sana will use is

A. SD card.
B. flash drive.
C. CD-ROM.
D. DVD-ROM.

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19. An example of sequential storage device is

A. DVD.
B. flash drive.
C. magnetic tape.
D. magnetic disk.

20. Which of the following is a component of a central processing unit?

A. Expansion cards
B. Motherboard
C. Control unit
D. System unit

21. Consider the given table that shows four memory addresses and the content stored in them.

Memory Address	Content
00100	10101010
00101	11001100
00110	11110000
00111	00110011

If a program counter (PC) holds the address 00101 at the beginning of the fetch cycle, then the content stored in the given registers would be

	Memory Address Register	Instruction Register
A	00101	11001100
B	00110	11110000
C	00111	00110011
D	00100	10101010

22. The type of CPU instruction that is responsible for changing the sequence of program execution, such as branching to different program segments is

A. data transfer instruction.
B. input/ output instruction.
C. data processing instruction.
D. program control instruction.

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23. The instruction format that allows the operation to be performed on one explicit source operand, and the result to be stored in a second explicit destination operand is

A. zero address instruction.
B. one address instruction.
C. two address instruction.
D. three address instruction.

24. A stage of fetch-decode-execute cycle, that begins when the instruction is received by the control unit is

A. store.
B. fetch.
C. decode.
D. execute.

25. CISC architecture focuses on providing a wide variety of instructions to handle complex tasks efficiently, often involving memory access.

In contrary to the above statement, RISC architectures simplifies instructions and places a heavier emphasis on

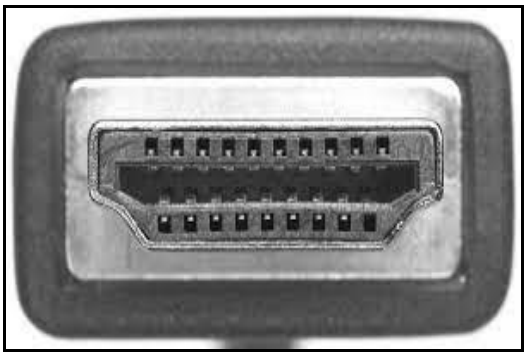
A. including instructions for arithmetic calculations only.
B. reducing the complexity of individual instructions.
C. reducing the number of available instructions.
D. minimising the number of registers available.

26. The PRIMARY function of the power supply within the system unit is to

A. stores electrical power.
B. control input and output devices.
C. process data and perform calculations.
D. supply electrical power to all internal components.

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27. Given is the image of a computer port.



The name of this port is a/ an

A. PS/2 port.
B. serial port.
C. HDMI port.
D. fire wire port.

28. The expansion card that can provide the data transfer speed up to 1000 Mbps is

A. gigabit card.
B. modem card.
C. wireless network card.
D. network interface card (NIC).

29. Which of the following options CORRECTLY describes the memory chip allocation on both the sides of SIMM and DIMM?

	Memory chip allocation on SIMM	**Memory chip allocation on DIMM**
A	Yes	Yes
B	Yes	No
C	No	Yes
D	No	No

30. If Tooba wants to connect her fitness tracker device with the smart phone, then the type of network she will use is

A. wide area network.
B. local area network.
C. near-me-area network.
D. personal area network.

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31. In a network, the component that is responsible for initiating the communication process by creating and sending data or information is

A. sender.
B. medium.
C. protocol.
D. receiver.

32. In fibre optic cables, the data is transmitted through

A. light waves.
B. radio waves.
C. sound waves.
D. electromagnetic waves.

33. The number of layers in TCP/ IP model is

A. two.
B. four.
C. six.
D. seven.

34. An IPv4 address is made up of

A. 4 bits.
B. 8 bits.
C. 4 bytes.
D. 8 bytes.

35. An example of wireless Local Area Network (WLAN) is

A. Wi-Fi.
B. bluetooth.
C. cellular network.
D. satellite communication.

36. In contrast to wired network, wireless network has

A. greater security.
B. mobility and flexibility.
C. lower initial setup costs.
D. higher data transfer speeds.

37. The PRIMARY purpose of a SIM (Subscriber Identity Module) card in mobile communication is to

A. authenticate the user and connect to the mobile network.
B. play multimedia content using mobile network.
C. store contact information in the mobile device.
D. provide electrical power to the mobile device.

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38. The component that is responsible for routing incoming and outgoing calls in a mobile communication network is

A. base station.
B. mobile phone.
C. switching node.
D. landline telephone network.

39. All of the following are common applications of GPS technology EXCEPT

A. weather forecasting.
B. navigation and mapping.
C. geocaching and outdoor activities.
D. vehicle tracking and fleet management.

40. Usman is working on a project that involves storing geographical data, including maps, coordinates and related information.

The MOST suitable database model for this project will be

A. network database.
B. relational database.
C. hierarchical database.
D. object-oriented database.

41. Consider the given tasks.

I. Inserting data into the database
II. Deleting objects from the database
III. Deleting records from the database
IV. Granting permissions to the database users

The task(s) performed by data control language of SQL is/ are

A. I only.
B. IV only.
C. I, II and III.
D. II, III and IV.

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42. The name of data type used in the field Married is

ID	FirstName	Married
1	Omar	☑
2	Fatima	☐
3	Muneeb	☐
4	Wajeeha	☑

A. Text.
B. Yes/ No.
C. Number.
D. OLE Object.

43. A customer database in an e-commerce platform uses email addresses to uniquely identify users. However, the platform also assigns a user-specific ID number.

What type of key is the user-specific ID number?

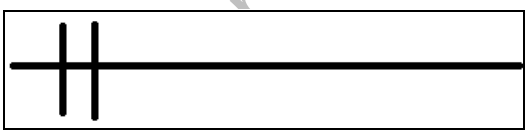
A. Secondary Key
B. Alternate Key
C. Primary Key
D. Foreign Key

44. In a hospital management system database, there are two entities: "Doctor" and "Patient." A doctor can treat multiple patients, and a patient can consult multiple doctors.

The type of relationship exists between the "Doctor" and "Patient" entities is

A. one-to-one relationship.
B. one-to-many relationship.
C. many-to-one relationship.
D. many-to-many relationship.

45. Consider the given image representing cardinality and modality.



The CORRECT interpretation of cardinality and modality from the given image is that

A. cardinality is one and modality is one.
B. cardinality is zero and modality is zero.
C. cardinality is zero and modality is many.
D. cardinality is many and modality is many.

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46. In an MS Access database, if you want to retrieve the "ProductDescription" and "Price" fields from a table called "Inventory", the CORRECT query will be

    A. Select ProductDescription, Price From Inventory;
    B. Select *ProductDescription, *Price From Inventory;
    C. Select ProductDescription AND Price From Inventory;
    D. Select *(ProductDescription AND Price) From Inventory;

47. In a database table named "Employees," you need to fetch records sorted by their "Salary" column from low to high salary.

    The SQL (DML) statement used will be

    A. Select * From Employees Sort Salary ASC;
    B. Select * From Employees Order Salary ASC;
    C. Select * From Employees Sort By Salary ASC;
    D. Select * From Employees Order By Salary ASC;

48. In a database, there is a table named "Inventory" that needs to be deleted completely.

    The SQL (DDL) statement used for this purpose will be

    A. Truncate Table Inventory;
    B. Remove Table Inventory;
    C. Delete Table Inventory;
    D. Drop Table Inventory;

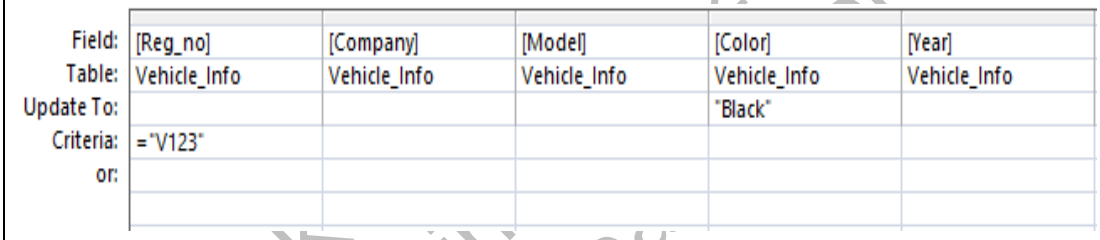
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Use the given database table named as Vehicle to answer Q.49 and Q.50.

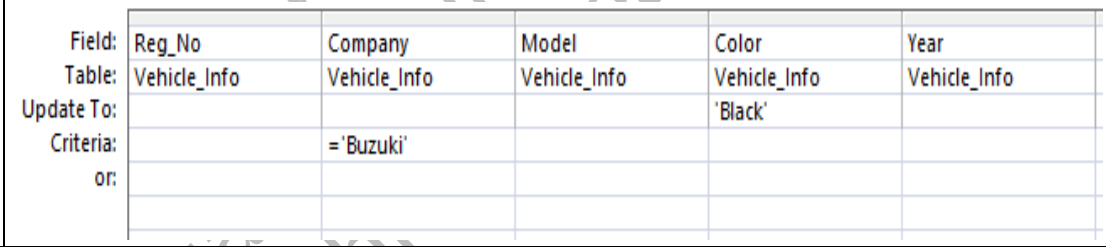
Reg_No	Company	Model	Color	Year
V121	Fonda	Rezel	White	2019
V122	Photon	Gaga	Black	2022
V123	Buzuki	Gultus	Silver	2021
V124	Toyoto	Norolla	Black	2020
V125	Buzuki	Diaz	Grey	2019

49. The query created in design view to change the color of Buzuki Gultus to Black from Silver is

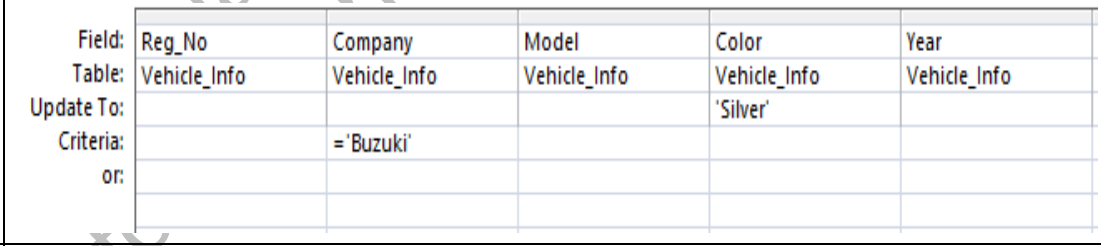
A



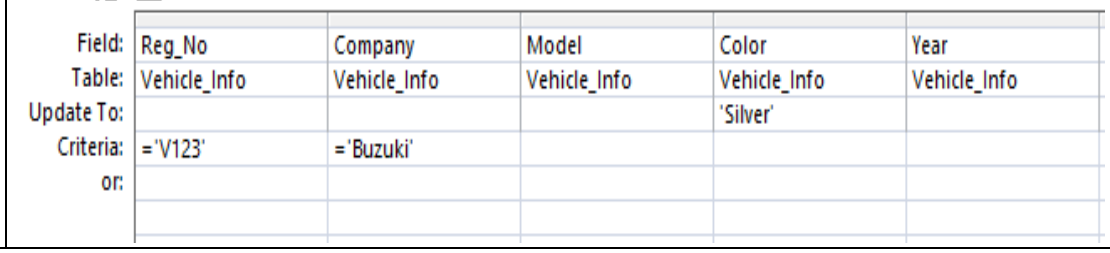
B



C

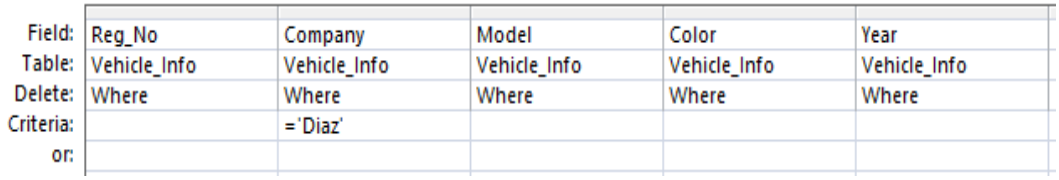


D



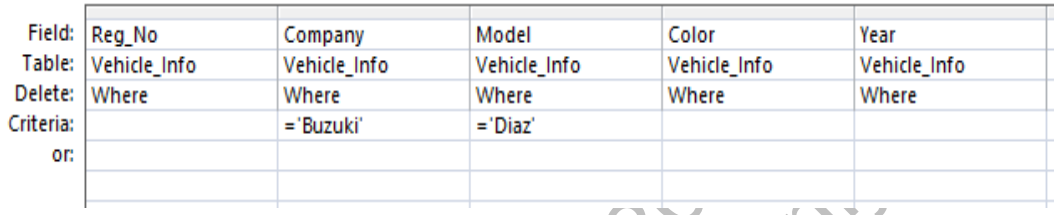
50. The query created in design view to delete an existing record of Buzuki Diaz is

A



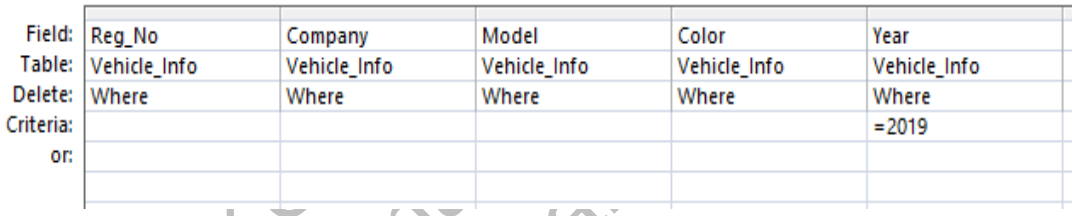
Field:	Reg_No	Company	Model	Color	Year
Table:	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info
Delete:	Where	Where	Where	Where	Where
Criteria:		='Diaz'			
or:					

B



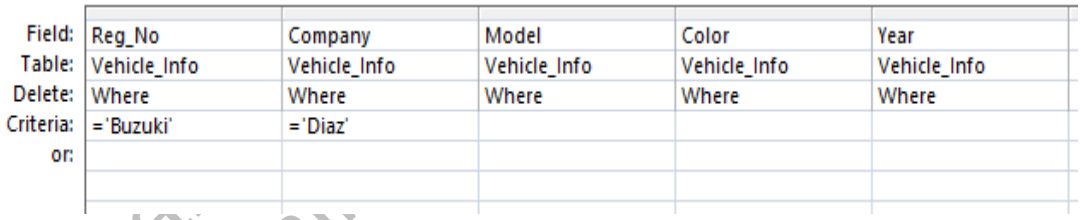
Field:	Reg_No	Company	Model	Color	Year
Table:	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info
Delete:	Where	Where	Where	Where	Where
Criteria:		='Buzuki'	='Diaz'		
or:					

C



Field:	Reg_No	Company	Model	Color	Year
Table:	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info
Delete:	Where	Where	Where	Where	Where
Criteria:					=2019
or:					

D



Field:	Reg_No	Company	Model	Color	Year
Table:	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info	Vehicle_Info
Delete:	Where	Where	Where	Where	Where
Criteria:	='Buzuki'	='Diaz'			
or:					

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